

Marcus E. Madsen*

Department of Mechanical Engineering,
Brigham Young University,
Provo, UT 84602, USA

Audrey K. Christiansen*

Department of Mechanical Engineering,
Brigham Young University,
Provo, UT 84602, USA

Jinkwan Han*

Department of Mechanical Engineering,
Brigham Young University,
Provo, UT 84602, USA

Jeffrey R. Hill^{†1}

Department of Mechanical Engineering,
Brigham Young University,
Provo, UT 84602, USA
email: jeff.hill@byu.edu

Sterling G. Baird^{†1}

Department of Mechanical Engineering,
Brigham Young University,
Provo, UT 84602, USA
email: sterling.baird@byu.edu

Bayesian Optimization of Multi-Material 3D-Printed Tensegrity-Inspired Structures for Energy Absorption

Tensegrity-inspired architectures combine rigid compression members with a continuous network of soft tension elements, yielding lightweight assemblies with tunable nonlinear responses that are attractive for energy-absorbing applications such as protective equipment, packaging, and aerospace landing structures. Multi-material fused-deposition modeling (FDM) of polylactic acid (PLA) struts and thermoplastic polyurethane (TPU) tension elements makes it possible to fabricate diverse tensegrity-inspired geometries rapidly enough to support a high-throughput experimental search, while the rate-dependent damping of TPU is itself an asset for impact mitigation. We report an experiment-driven Bayesian optimization (BO) framework in which a Gaussian-process surrogate is trained directly on physical quasi-static compression and drop-weight impact data and used to recommend the next batch of designs to print and test. The framework parameterizes a family of tensegrity-inspired unit cells over strut geometry, tension-element cross-section, connectivity topology, and unit-cell tiling, and optimizes peak transmitted force, specific energy absorption, and compaction efficiency.

Keywords: tensegrity, multi-material 3D printing, Bayesian optimization, energy absorption, design of experiments

xxiii
xxiv
xxv
xxvi
xxvii
xxviii
xxix
xxx
xxxi
xxxii
xxxiii
xxxiv
xxxv
xxxvi
xxxvii
xxxviii
xxxix
xl
xli
xlii

1 Introduction

Tensegrity structures are assemblies of rigid compression members suspended within a continuous network of tension members; their defining property, that no two rigid bars touch, yields lightweight assemblies with remarkable strength-to-weight ratios and tunable nonlinear mechanical responses [1,2]. These properties have made tensegrity attractive across scales, from metamaterial unit cells [3–5] to compliant impact-absorbing robots developed for planetary landing [6–10]. Recent work has shown that tensegrity-inspired unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact [11], opening a path toward additively manufactured architected absorbers. Throughout, we use *tensegrity-inspired* to denote printable unit cells that adopt the characteristic tensegrity motif—rigid compression members carried by a network of pretensioned soft tension elements—while relaxing the strict classical requirement that no two compression members touch and that the tension members behave as ideal inextensible cables; here the tension elements are extruded TPU and the struts may share printed junctions.

Multi-material fused-deposition modeling (FDM) that pairs a rigid polymer such as polylactic acid (PLA) with a soft, rate-dependent elastomer such as thermoplastic polyurethane (TPU) has been used to fabricate rigid–soft architected energy absorbers on a single platform—for example, thick-panel origami metamaterials with PLA panels and TPU hinges [12] and ABS/TPU honeycombs with tunable stiff/flexible proportions [13]. The same rigid–soft co-printing capability makes it possible to rapidly fabricate diverse tensegrity-inspired geometries, which adopt a discrete-compression/continuous-tension topology rather than an origami or honeycomb one. Although extruded TPU does not behave as an idealized inextensible cable, its rate-dependent damping is a desirable property for impact-absorption applications and complements the elastic stiffness of the PLA compression members rather than

competing with it.

Designing such architected absorbers is fundamentally a *design-of-experiments* problem: the configuration space spanned by strut geometry, tension-element cross-section, connectivity, and unit-cell tiling is large; each candidate must be physically printed and tested; and the relevant performance metrics—peak transmitted force, specific energy absorption (SEA), and compaction efficiency—are noisy and partially correlated. Bayesian optimization (BO) is well-matched to this regime, having driven breakthroughs from hyperparameter tuning of AlphaGo [14] to sustainable concrete [15], autonomous discovery of organic laser emitters [16], and architected materials [17–20].

1.0.1 Contributions. This paper makes the following contributions:

- (1) A parameterized family of multi-material 3D-printable tensegrity-inspired unit cells with PLA compression members and TPU tension elements that are anchored *inside* the ends of each PLA strut—the strut acting as a rigid cage in which the cables meeting at a given end join before exiting through discrete outlets—to ensure cyclic interface durability.
- (2) An experiment-driven BO loop, built on BoTorch/Ax [21,22], that operates directly on physical quasi-static compression and drop-weight impact measurements, recommending the next designs to fabricate.
- (3) High-fidelity validation experiments using *pretensioned* tensegrity assemblies (true cables and measured pretension) for the Pareto-optimal designs identified by the single-stage loop above, reported separately rather than fused into the optimization.

The remainder of the paper is organized as follows. Section 2 reviews tensegrity mechanics, multi-material 3D printing for energy absorption, and Bayesian optimization with an emphasis on multi-objective and noise-robust acquisition functions. Section 3

34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66

[†]Corresponding Author.
June 11, 2026

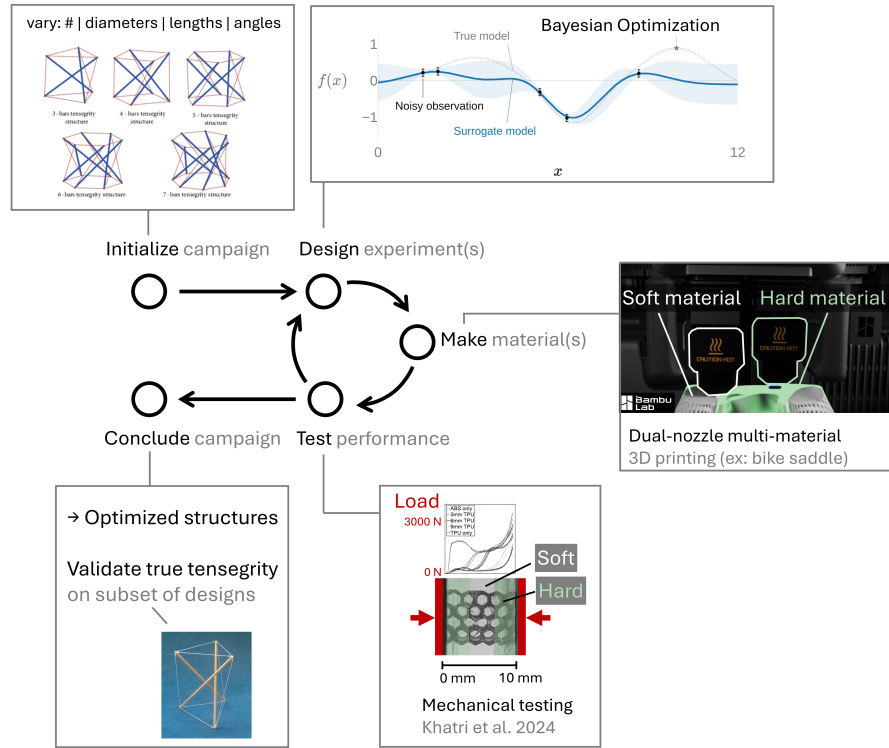


Fig. 1 Closed-loop, experiment-driven design framework. Parameterized PLA/TPU tensegrity-inspired unit cells are printed, tested under quasi-static compression and drop-weight impact, and the resulting performance data drive a Gaussian-process surrogate that recommends the next batch of designs to fabricate.

introduces the design parameterization, fabrication workflow, experimental protocols, and BO loop. Section 4 reports the results of the closed-loop experimental campaign, and Section 5 discusses the implications and limitations of the approach. Section 6 concludes.

2 Background

2.1 Tensegrity-Inspired Architectures for Energy Absorption. Tensegrity prisms and their planar extensions exhibit softening–stiffening response under compression [3–5], characteristics desirable for impact mitigation because the structure can absorb a large fraction of the input energy at a controlled, sub-injurious force plateau. Santos [23] recently characterized a tensegrity energy-dissipation metamaterial reporting equivalent viscous damping on the order of ~23% with negligible residual strain per impact, underscoring the suitability of tensegrity-inspired architectures for repeated-impact use cases. The space program literature documents large-scale tensegrity-inspired absorbers for planetary landing [6–10] and a long history of crushable energy-absorbing landing systems [24–26]. Pajunen et al. [11] demonstrated that truncated-tetrahedral tensegrity-inspired unit cells can be directly 3D-printed and exhibit load-limiting plateaus, motivating the present focus on additively manufactured TPU/PLA composites. More recent work on dynamically loaded additively-manufactured energy-absorbing lattices [27], together with Intrigila et al.’s bistable tensegrity-like unit cell for lattice metamaterials [28] as the closest published analog, confirms that this architecture class also responds favorably under impact and quasi-static loading, although neither study employed multi-material FFF or closed-loop design optimization.

2.2 Multi-Material 3D Printing of PLA/TPU Composites. Multi-material rigid–soft FDM enables stiff/compliant combina-

tions on a single platform with tunable energy absorption, demonstrated for thick-panel origami with PLA (or ABS/CFRP) panels and TPU hinges [12] and for ABS–TPU honeycombs [13]. Neither study is itself a tensegrity, but both establish the rigid–soft co-printing route we adopt. Ye et al. [12] introduced a *wrapping-based* strategy in which rigid cores are encapsulated by continuous soft skins, preventing interface delamination and enabling cyclic durability, something we take inspiration from in our designs. Recent multi-material PLA/TPU sandwich and layered composites [29,30] report quantitative bounds on stiffness, strength, and energy absorption for the same PLA–TPU pair used here, while fused filament fabrication (FFF) PLA fatigue data [31,32] inform conservative design bounds for the rigid struts. Architected multi-material lattices with co-printed compliant skins have been explicitly designed for high specific energy absorption [33], with FFF-specific multimaterial interface adhesion (notably TPU bonded to rigid co-printed substrates) characterized in [34]; recent surveys catalogue the rapidly expanding additively manufactured polymeric energy-absorption literature [35,36].

2.3 Bayesian Optimization for Architected-Material Design. Bayesian optimization is a sequential, model-based strategy for optimizing expensive black-box functions using a Gaussian-process surrogate [37,38]. Modern implementations such as BoTorch [21] and the Ax platform built atop it [22] support parallel batched evaluations and a wide variety of acquisition functions; recent algorithmic advances include log-EI for numerical robustness [39], noisy expected hypervolume improvement for multi-objective settings [40], input-noise-robust acquisitions [41], and evolution-guided constrained multi-objective BO for self-driving labs [42]. Mixed/categorical search spaces—needed here for the discrete connectivity-topology variable—have dedicated treatments in BOCS [43] and the one-hot-with-rounded-kernel construction of [44]. BO has been applied to mechanical metamaterial

and structural design [17–19] and to additive manufacturing [20]. Mo et al. [17] in particular established a multifidelity BO framework for architected materials by fusing low-fidelity simulations with high-fidelity ground truth via nonlinear information-fusion priors [45]; the present work shifts the emphasis from simulation surrogates to direct experimental data, appropriate for materials whose constitutive response is difficult to calibrate from first principles.

3 Materials and Methods

3.1 Design Parameterization. We parameterize a family of tensegrity-inspired unit cells via four groups of design variables:

- **PLA compression members.** Strut diameter d_s and length ℓ_s , with the slenderness ratio ℓ_s/d_s bounded away from buckling-dominated regimes.
- **TPU tension elements.** Cross-sectional area A_t (or equivalent diameter d_t for circular cross-sections) and pre-tension geometry imposed during printing.
- **Connectivity topology.** Selected from a discrete set of candidate unit cells (e.g., truncated-tetrahedral, prism-based, and octahedral variants); the design vector encodes a categorical choice.
- **Unit-cell tiling.** Number of cells $N_x \times N_y \times N_z$ and the orientation of each cell relative to the impact direction.

3.1.1 Working prototype. The first instance of this family used to validate the fabrication and test pipeline is a three-strut (T_3) tensegrity prism scaled to a ~ 50 mm bounding box, with $d_s =$ and a cable diameter d_t swept over the FFF-resolvable set $\{1.2, 1.8, 2.4, 3.0, 4.5\}$ mm. Under the prism’s D_3 symmetry the twelve member-diameter axes collapse to four orbit axes (one strut orbit and three cable orbits: saddle, top, bottom), which keeps the initial design vector low-dimensional while leaving room to relax to per-member diameters once the saddle/top/bottom orbits have been individually characterized. [Figure 2 shows the parametric CAD model alongside an as-printed prototype of this working \$T_3\$ prism.](#)

Table 1 Design variables for the T_3 -prism Sobol initialization batch, as implemented in the BO harness (single-iteration $n = 9$ Sobol set). The joint diameter is held fixed at the `t3-prism.scad` default; all specimens are printed in the vertical orientation and packed 3×3 on a single Bambu Lab H2D plate.

Design variable	Symbol	Units	Range
Base radius	R	mm	25–40
Cell height	H	mm	60–110
Twist angle	θ	deg	40–80
Strut (PLA) diameter	d_s	mm	6.0–12.0
Cable (TPU) diameter	d_t	mm	3.0–5.5
Joint diameter (fixed)	d_j	mm	7.0

3.2 Multi-Material Fabrication. Specimens are fabricated on a multi-material FDM printer using PLA for the rigid struts and TPU for the soft tension elements. [Figure 3 summarizes the end-to-end fabrication and characterization workflow, from parametric CAD through multi-material slicing, dual-extrusion printing, post-processing, and mechanical testing.](#) Rather than wrapping the struts in a continuous TPU skin, the current design anchors the TPU tension elements *inside* the ends of each PLA strut: the cables converging at a given strut end are joined within the strut, which acts as a rigid cage housing the junction, and the individual tendons then exit through discrete outlets. This keeps the soft–rigid interface in internal, compression-dominated pockets rather than at an exposed overmolded surface. This construction

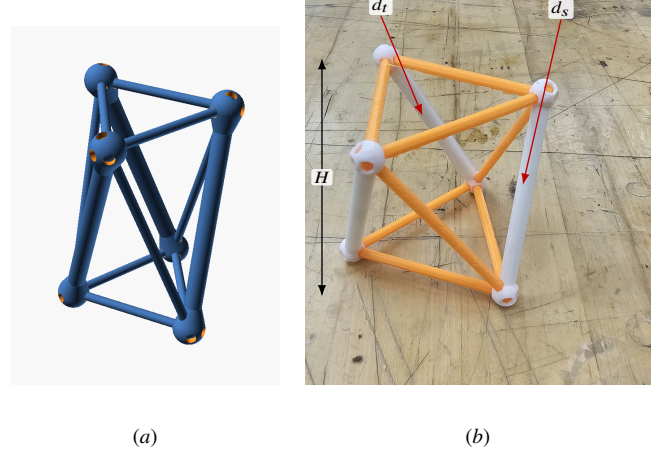


Fig. 2 Working T_3 -prism prototype. (a) Parametric OpenSCAD model whose base radius R , cell height H , twist angle θ , strut diameter d_s , and cable diameter d_t are the design variables of Table 1. (b) A representative dual-material print from the PR #35 print history (rigid PLA struts, here in white; tendon members in orange), with callouts marking the strut diameter d_s , a tendon (cable) diameter d_t , and the cell height H ; the inter-plate twist θ is the relative rotation of the top and bottom node triangles, and the base radius R sets their circumradius.

effectively inverts the material roles of Ye et al.’s soft-skin-over-rigid-core wrapping [12]: here a soft TPU core is captured within a rigid PLA shell, so we cite that work only as a multi-material co-printing precedent rather than as validation of the present junction.

3.2.1 Print platform and joint geometry. The working platform is a Bambu Lab H2D printer, which permits a single rigid-soft build without filament swaps. Strut endpoints are tied to cables through parametric joint geometries developed and ranked through a five-design OpenSCAD study (anchor-bulb, dovetail, TPU-sleeve overmold, eyelet-loop, and TPU-rebar variants); the working prototype uses a dovetail joint (Design B) with an anchor-bulb backup (Design A) for sensitivity studies, with a captive TPU core routed inside the PLA shell to keep the soft tendon protected from layer-line failure at the strut-to-cable transition. The full five-design comparison and the selected junction are detailed in the Supplementary Information.

3.2.2 Supports for soft members. Because near-vertical TPU tendons are otherwise unsupported during printing, the slicer profile combines a tensegrity-specific Bambu Studio recipe (support threshold angle dropped from 40° to 10° , support material matched to the rigid extruder) with manually generated narrowing pillars that ray-cast against the printable mesh underside, producing TPU-safe coverage of vertical members without slicer-side painting. Full workflow details are provided in the Supplementary Information.

3.3 Experimental Characterization. Each printed specimen is subjected to two physical tests:

3.3.1 Quasi-static compression. A uniaxial load–displacement curve is recorded on a benchtop electromechanical load frame (BYU CB 123 Instron) following best-practice combinations of ASTM D638, ASTM D412, ASTM E111, ASTM F2971, and ISO/ASTM 52900/52921 adapted to architected polymer cells. Because expected cell stiffnesses fall in the ~ 5 – 30 g force range, the default 5 kN load cell is over-ranged and is being replaced by a 100–500 N cell for the first campaign.

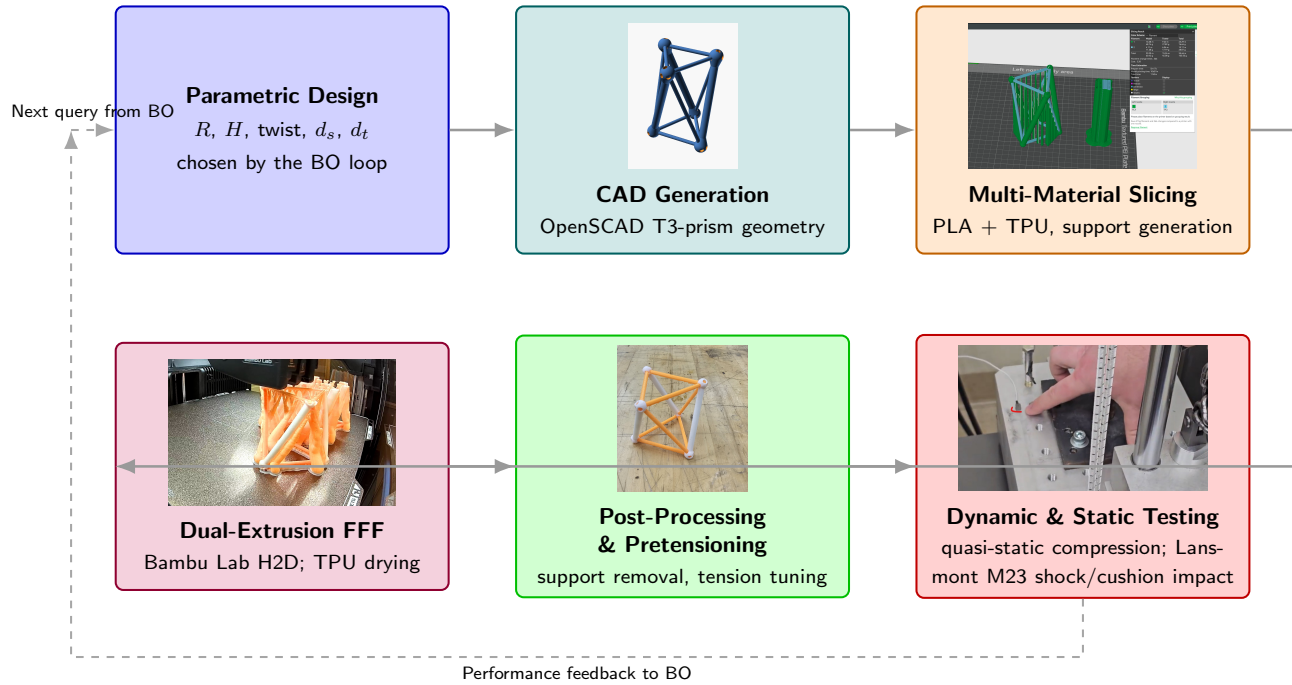


Fig. 3 Fabrication and characterization workflow for the multi-material tensegrity-inspired unit cells: design parameters drive parametric CAD, slicing with manually generated TPU supports, a single multi-material print on the Bambu Lab H2D, post-processing and inspection, and mechanical testing. Node photographs are representative artifacts drawn from the project's build history: the OpenSCAD T_3 -prism render (CAD), the dual-nozzle Bambu Studio slice with PLA on the left extruder and TPU on the right (slicing), an in-progress H2D print of the prism batch (FFF), an as-printed specimen (post-processing), and the bungee-assisted drop-tower base plate with accelerometer (testing).

Table 2 Multi-material print parameters on the Bambu Lab H2D. Cells marked *TBD* are pending confirmation; the reporting structure follows the methods section of Ye et al. [12].

Parameter	PLA (rigid struts)	TPU (soft tendons)
Nozzle temperature (°C)	<i>TBD</i>	<i>TBD</i>
Bed temperature (°C)		<i>TBD</i>
Layer height (mm)		<i>TBD</i>
Print speed (mm/min)	<i>TBD</i>	<i>TBD</i>
Infill (%)	<i>TBD</i>	<i>TBD</i>
Build-plate type		Textured PEI
Build orientation		Vertical
Supports	Off in slicer; manual narrowing pillars	

214 Reported metrics are

$$215 \quad \text{SEA} = \frac{1}{m} \int_0^{\delta_d} F(\delta) d\delta, \quad \eta_c = \frac{\int_0^{\delta_d} F(\delta) d\delta}{F_{\max} \delta_d}, \quad (1)$$

216 where m is the specimen mass, δ_d is the densification displacement, $F_{\max}$ is the peak transmitted force, and η_c is the compaction efficiency. To keep these metrics objective and reproducible, the densification displacement δ_d is taken as the displacement at which the energy-absorption efficiency–displacement curve reaches its maximum ($d\eta/d\delta = 0$), and $F_{\max}$ is the maximum force recorded within the integration window $[0, \delta_d]$ rather than a global or post-densification peak.

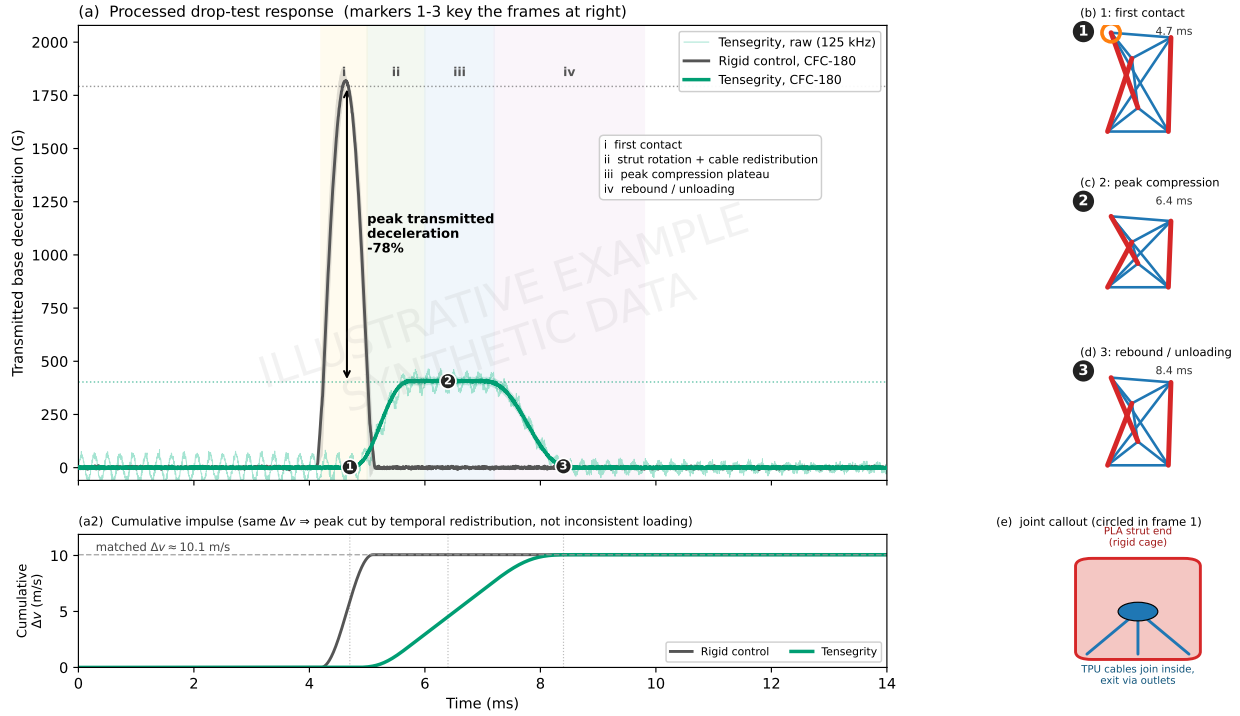
224 **3.3.2 Drop-weight impact.** Following Pajunen et al. [11], specimens are subjected to drop-weight impact tests with instrumented load cells; we report the peak transmitted force and the energy-absorption plateau characteristics. Figure 4 illustrates the

mechanism-oriented presentation we adopt for these processed deceleration traces, linking features of the measured signal to the structural events that produce them. The primary fixture is a bungee-assisted laboratory drop tower in which the base accelerates downward faster than 1g; because unconstrained specimens lift off the base during descent by design of the rig, the protocol constrains the specimen top through light tethers (capping upward travel) and registers the base via transfer tape or V-block features so that loading-direction compliance is not altered. A higher-fidelity replicate campaign on a Lansmont M23 shock/cushion tester (per ASTM D1596) with synchronized Polytec QTEC laser vibrometry is planned for selected Pareto-optimal designs. Slip resistance and traction at the specimen–floor interface—governed by the tip geometry rather than the internal lattice—follow the test methodology of ISO 11334-4 [46] for walking-aid tips and are out of scope for the present axial-impact study.

244 **3.3.3 Per-modality objectives.** Five complementary data streams (triaxial accelerometer, high-speed video, shaker transfer function, slug-firing gas gun, and Polytec QTEC laser-Doppler vibrometer) each contribute a distinct raw observable; the mapping from each stream to BO-ready scalar objectives (SEA, VEA, $F_{\max}$, cushion-curve intercept, Gibson–Ashby relative density) is defined in five companion briefs and consolidated in a cross-modality synthesis.

252 **3.4 Experiment-Driven Bayesian Optimization Loop.** The optimization loop maintains a Gaussian-process (GP) surrogate over the design space defined in Section 3.1. After each round of physical experiments, the GP is updated and an acquisition function is maximized to recommend the next batch of designs to fabricate. Implementation uses BoTorch [21] with the log-Expected-Improvement (LogEI) acquisition [39] for the single-objective baseline and noisy Expected Hypervolume Improvement

EXAMPLE (mock-up, synthetic data): mechanism-oriented drop-test figure



Synthetic illustration. In a real figure: transmitted base deceleration of a rigid control vs. T3-prism tensegrity, both SAE J211 CFC-180 filtered (125 kHz raw shown faint), zero-phase (filtfilt), time-aligned on the CH4 first-contact index. Curves are the mean of $n=5$ replicates; bands are ± 1 s.d. Frames 1-3 are high-speed/DIC stills at markers 1-3; (e) shows the internal cable-anchor joint that spreads the impulse.

Fig. 4 Illustrative example (synthetic data). Template for the mechanism-oriented drop-test figures planned for the Results: (a) processed deceleration response of a rigid control vs. a tensegrity specimen (SAE J211 CFC-180 filtered), with event-based phase shading and numbered markers keyed to the schematic frames (b)–(d); (a2) cumulative-impulse subpanel showing the two traces transfer the same mass-normalized Δv , so the lower tensegrity peak reflects temporal redistribution rather than inconsistent loading; (e) the internal-anchor joint load path. Curves are synthetic and impulse-consistent (control anchored to the documented ~1792 G CFC-180 peak; tensegrity shoulder scaled to the same impulse), pending real high-rate channels and camera frames. Generated by scripts/figures/mechanistic_data_figure_example.py.

(qNEHVI) [40] for the multi-objective case (simultaneously maximizing SEA and compaction efficiency while constraining peak transmitted force). When experimental noise dominates (e.g., between-print variability), we additionally evaluate the input-noise-robust acquisition of Daulton et al. [41] and the evolution-guided constrained variant of Low et al. [42]. The discrete connectivity-topology variable is handled with the mixed-search-space construction of [44]; the BOCS combinatorial baseline [43] is reported for comparison.

3.4.1 Implementation status. The campaign harness is built on Ax's HierarchicalSearchSpace so that printer-feasibility constraints (e.g. the cable-diameter categorical depending on the chosen connectivity topology) can be encoded directly in the search space rather than enforced post hoc. The first physical batch consists of $n = 9$ specimens generated by a Sobol sequence over the four D_3 -orbit axes of the working T_3 prism and printed in a single 3×3 plate layout. For the present ≤ 25 -dimensional parameterization the default acquisition pairing is sparse axis-aligned subspace BO (SAASBO) with qNEHVI; escalation to trust-region BO (TuRBO) is planned only if larger per-member-diameter tilings push the design space well beyond this regime.

Once the campaign produces model data, the fitted surrogate is audited with the leave-one-out cross-validation and parameter-sensitivity diagnostics of Figs. 5 and 6.

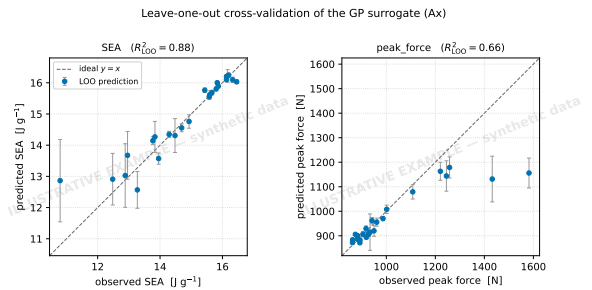


Fig. 5 Illustrative example (synthetic data). Leave-one-out cross-validation of the Gaussian-process surrogate—predicted vs. observed specific energy absorption (SEA) and peak transmitted force, with model error bars and R^2_{LOO} —obtained from a real Ax/BoTorch run on synthetic objective values. To be regenerated from campaign data.

4 Results

4.1 Convergence of the BO Loop. Figure 7 illustrates the intended convergence comparison.

4.2 Pareto-Optimal Designs. Figure 8 shows the form the Pareto-front presentation will take.

4.3 Reproducibility Across Print Replicates.

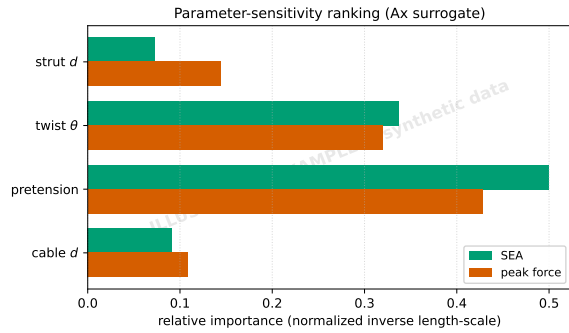


Fig. 6 *Illustrative example (synthetic data). Parameter-sensitivity ranking (model length-scale-based importances per objective; cf. SAASBO inverse length-scales or Sobol indices) of the design variables in Table 1, from the same Ax/BoTorch run as Fig. 5.*

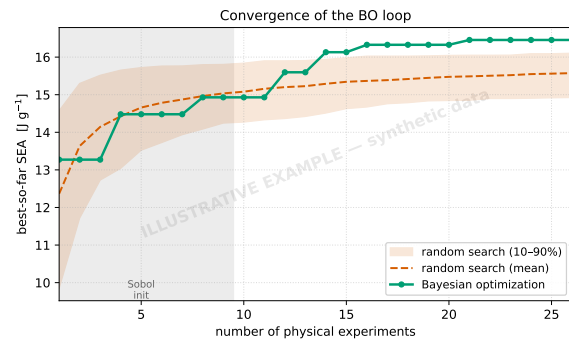


Fig. 7 *Illustrative example (synthetic data). Best-so-far SEA vs. number of physical experiments for the Ax/BoTorch loop against an independent random-search baseline; the shaded region is the $n = 9$ Sobol initialization. Generated on synthetic objective values; to be replaced with campaign results (with confidence intervals over random seeds, and Pareto hypervolume for the multi-objective runs).*

5 Discussion

Topics to address:

- Comparison to the simulation-driven multifidelity framework of Mo et al. [17] and the single-condition fabrication study of Pajunen et al. [11].
- Practical considerations: TPU–PLA interface durability under cyclic loading; sensitivity of optimal designs to print parameters (extrusion temperature, layer height); transferability to TPU/PETG or other high-temperature thermoplastics.
- Path forward toward integrated tread+lattice structures, miniaturization to crutch-tip envelopes (from the Edison crutch-tip synthesis), and field testing. Long-term crutch use is associated with substantial upper-extremity overuse injury and ground-reaction-force impulses [47–51]; existing spring-loaded [52,53] and polymer-damped [54] tip designs reduce loading rate but not peak transmitted force, leaving an open design opportunity that an architected tensegrity-inspired insert is well-positioned to address. Recent open-source 3D-printed crutch work [55] and expert evaluations of orthopaedic caps [56] suggest a plausible translation path; adjacent footwear and orthotic lattice studies [57–59] provide additional transferable design heuristics.

5.0.1 Limitations..

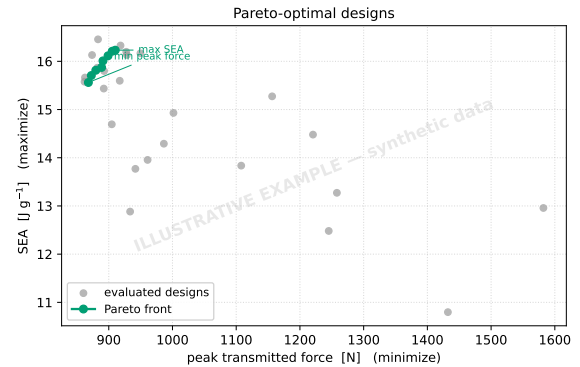


Fig. 8 *Illustrative example (synthetic data). Pareto front in (peak transmitted force, SEA) space with all evaluated designs, from the Ax/BoTorch loop on synthetic objective values; representative geometries to be annotated once campaign data are available.*

Table 3 Placeholder. Mean and CV of SEA, peak force, and compaction efficiency across n replicate prints for the top- k designs.

Column A	Column B	Column C
(table contents pending)		

6 Conclusions

We have presented an experiment-driven Bayesian-optimization framework for designing multi-material 3D-printed tensegrity-inspired energy absorbers using PLA struts and TPU tension elements. The framework parameterizes a family of unit cells, fabricates each candidate on a single multi-material FDM platform with TPU tendons anchored inside the PLA strut ends, characterizes the response under quasi-static compression and drop-weight impact, and updates a Gaussian-process surrogate [21] that recommends the next batch of designs. The approach is broadly applicable to architected absorbers whose constitutive response is difficult to calibrate from first principles and where physical experiments—rather than simulations—are the natural ground truth.

Acknowledgment

This work was supported by the Brigham Young University Mentored Research Grant program. The authors thank the undergraduate researchers who carried out the multi-material printing and impact testing, and the graduate co-mentor who supported day-to-day operations.

Funding Data

- Brigham Young University Mentored Research Grant ().

Conflict of Interest

The authors declare no conflict of interest.

References

- [1] Skelton, R. E. and de Oliveira, M. C., 2009, *Tensegrity Systems*, Springer, New York.
- [2] Sultan, C., 2009, “Tensegrity: 60 Years of Art, Science, and Engineering,” *Advances in Applied Mechanics*, **43**, pp. 69–145.
- [3] Amendola, A., Carpentieri, G., de Oliveira, M. C., Skelton, R. E., and Fraternali, F., 2014, “Experimental Investigation of the Softening–Stiffening Response of Tensegrity Prisms under Compressive Loading,” *Composite Structures*, **117**, pp. 234–243.

- [4] Zhang, Q., Zhang, D., Dobah, Y., Scarpa, F., Fraternali, F., and Skelton, R. E., 2018, "Tensegrity Cell Mechanism Metamaterial with Metal Rubber," *Applied Physics Letters*, **113**(3), p. 031906.
- [5] Fraternali, F., Carpentieri, G., and Amendola, A., 2015, "Tensegrity Ring Structures for Impact Protection," *International Journal of Solids and Structures*, **65**–**66**, pp. 232–243.
- [6] Agogino, A. K., SunSpiral, V., and Atkinson, D., 2018, "Super Ball Bot — Structures for Planetary Landing and Exploration," NASA Ames Research Center, <https://www.nasa.gov/general/super-ball-bot/>
- [7] Caluwaerts, K., Agogino, A. M., and SunSpiral, V., 2014, "SUPERball: Exploring Tensegrities for Planetary Probes," *Proceedings of the 12th International Symposium on Artificial Intelligence, Robotics and Automation in Space (i-SAIRAS)*.
- [8] Sabelhaus, A. P., Bruce, J., Caluwaerts, K., Manovi, P., Firoozi, R. F., Dobi, S., Agogino, A. M., and SunSpiral, V., 2015, "System Design and Locomotion of SUPERball, an Untethered Tensegrity Robot," *2015 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 2867–2873, doi: [10.1109/ICRA.2015.7139590](https://doi.org/10.1109/ICRA.2015.7139590).
- [9] Vespignani, M., Friesen, J. M., SunSpiral, V., and Bruce, J., 2018, "Design of SUPERball v2, a Compliant Tensegrity Robot for Absorbing Large Impacts," , pp. 2865–2871.
- [10] Deitrich, N., Baldonado, K. M., Khan, A., Cook, J., and Rizzo, L., 2022, "A Rideshare Tensegrity Rover Concept to Explore Titan's Lands and Oceans," *AIAA SciTech Forum*, NASA Langley Research Center / Ames Research Center.
- [11] Pajunen, K., Johanns, P., Pal, R. K., Rimoli, J. J., and Daraio, C., 2019, "Design and Impact Response of 3D-Printable Tensegrity-Inspired Structures," *Materials & Design*, **182**, p. 107966.
- [12] Ye, H., Liu, Q., Cheng, J., Li, H., Jian, B., Wang, R., Sun, Z., Lu, Y., and Ge, Q., 2023, "Multimaterial 3D Printed Self-Locking Thick-Panel Origami Metamaterials," *Nature Communications*, **14**, p. 1607.
- [13] Khatri, N. R. and Egan, P. F., 2024, "Energy Absorption of 3D Printed ABS and TPU Multimaterial Honeycomb Structures," *3D Printing and Additive Manufacturing*, **11**, pp. e840–e850.
- [14] Silver, D., Huang, A., Maddison, C. J., Guez, A., Sifre, L., van den Driessche, G., Schrittwieser, J., Antonoglou, I., Panneershelvam, V., Lanctot, M., et al., 2016, "Mastering the Game of Go with Deep Neural Networks and Tree Search," *Nature*, **529**(7587), pp. 484–489.
- [15] Ament, S., Witte, A., Garg, N., and Kusuma, J., 2023, "Sustainable Concrete via Bayesian Optimization," *arXiv preprint arXiv:2310.18288*, NeurIPS 2023 Workshop on Adaptive Experimental Design and Active Learning in the Real World.
- [16] Strieth-Kalthoff, F., Hao, H., Rathore, V., Derasp, J., Gaudin, T., Angello, N. H., Seifrid, M., Trushina, E., Guy, M., Liu, J., et al., 2024, "Decentralized, Asynchronous, Closed-Loop Discovery of Organic Laser Emitters," *Science*, **384**(6697), p. eadk9227.
- [17] Mo, C., Perdikaris, P., and Raney, J. R., 2023, "Accelerated Design of Architected Materials with Multifidelity Bayesian Optimization," *Journal of Engineering Mechanics*, **149**(6), p. 04023028.
- [18] Wang, Y. et al., 2022, "Bayesian Optimization for the Design of Mechanical Metamaterials," *Journal of the Mechanics and Physics of Solids*, **159**, p. 104734.
- [19] Lee, S. et al., 2023, "Bayesian Optimization for Structural Design Problems," *Structural and Multidisciplinary Optimization*, **66**(2), p. 47.
- [20] Zhang, M., Parnell, A., Brabazon, A., and Benavoli, A., 2021, "Bayesian Optimisation for Sequential Experimental Design with Applications in Additive Manufacturing," *arXiv preprint arXiv:2107.12809*.
- [21] Balandat, M., Karrer, B., Jiang, D. R., Daulton, S., Letham, B., Wilson, A. G., and Bakshy, E., 2020, "BoTorch: A Framework for Efficient Monte-Carlo Bayesian Optimization," *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, <https://arxiv.org/abs/1910.06403>
- [22] Olson, M., Santorella, E., Tiao, L. C., Cakmak, S., Garrard, M., Daulton, S., Lin, Z. J., Ament, S., Beckerman, B., Onofrey, E., Igusti, P., Lara, C., Letham, B., Cardoso, C., Shen, S., Lin, A. C., Grange, M., Kashtelyan, E., Eriksson, D., Balandat, M., and Bakshy, E., 2025, "Ax: A Platform for Adaptive Experimentation," *Proceedings of the Fourth International Conference on Automated Machine Learning*, L. Akoglu, C. Doerr, J. N. van Rijn, R. Garnett, and J. R. Gardner, eds., Proceedings of Machine Learning Research, Vol. 293, PMLR, pp. 21/1–25, <https://proceedings.mlr.press/v293/olson25a.html>
- [23] Santos, F. A., 2023, "Toward a novel energy-dissipation metamaterial with tensegrity architecture," *Advanced Materials*, May 2023.
- [24] Cloutier, G. J., 1966, "Landing Impact Energy Absorption Using Anisotropic Crushable Materials," *Journal of Spacecraft and Rockets*, **3**(12), pp. 1755–1761.
- [25] Adams, D. S., 2004, "Mars Exploration Rover Airbag Landing Loads Testing and Analysis," *45th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference*, doi: [10.2514/6.2004-1795](https://doi.org/10.2514/6.2004-1795).
- [26] Jackson, K. E., Fasanella, E. L., and Polanco, M. A., 2014, "Simulating the Response of a Composite Honeycomb Energy Absorber: I. Dynamic Crushing of Components and Multiterrain Impacts," *Journal of Aerospace Engineering*, **27**(3), pp. 424–436.
- [27] Davami, K., Mohsenizadeh, M., Munther, M., Palma, T., Beheshti, A., and Momeni, K., 2019, "Dynamic energy absorption characteristics of additively-manufactured shape-recovering lattice structures," doi: [10.1088/2053-1591/aaf78c](https://doi.org/10.1088/2053-1591/aaf78c), <https://doi.org/10.1088/2053-1591/aaf78c>
- [28] Intrigila, C., Micheletti, A., Nodargi, N. A., Artioli, E., and Bisegna, P., 2022, "Fabrication and experimental characterisation of a bistable tensegrity-like unit for lattice metamaterials," *Additive Manufacturing*, **57**, p. 102946.
- [29] Arifvianto, B., Satiti, B. E., Salim, U. A., Suyitno, Nuryanti, A., and Mahardika, M., 2022, "Mechanical properties of the fff sandwich-structured parts made of pla/tpu multi-material," *Progress in Additive Manufacturing*, **7**:1213–1223, Apr 2022.
- [30] Ruwais, A. and Naveed, N., 2025, "Mechanical performance of layered pla-tpu composites using multi-material additive manufacturing," *Unknown journal*, 2025.
- [31] Ezech, O. H. and Susmel, L., 2018, "On the fatigue strength of 3d-printed polylactide (pla)," *Procedia structural integrity*, **9**:29–36, Jan 2018.
- [32] Vanaei, H. R., Shirinbayan, M., Vanaei, S., Fitoussi, J., Khelladi, S., and Tcharkhtchi, A., 2021, "Multi-scale damage analysis and fatigue behavior of pla manufactured by fused deposition modeling (fdm)," *Rapid Prototyping Journal*, **27**:371–378, Jan 2021.
- [33] Yavas, D., Liu, Q., Zhang, Z., and Wu, D., 2022, "Design and fabrication of architected multi-material lattices with tunable stiffness, strength, and energy absorption," *Materials & Design*, **217**:110613, May 2022.
- [34] Caminero, M. A., Chacón, J. M., García-Plaza, E., Núñez, P. J., Reverte, J. M., and Becar, J. P., 2019, "Additive Manufacturing of PLA-Based Composites Using Fused Filament Fabrication: Effect of Graphene Nanoplatelet Reinforcement on Mechanical Properties, Dimensional Accuracy and Texture," *Polymers*, **11**(5), p. 799.
- [35] Bustihan, A. and Botiz, I., 2026, "Recent advances in additively manufactured polymeric structures for mechanical energy absorption," *Polymers*, **18**:1019, Apr 2026.
- [36] Liu, J., Seidu, R., Hu, L., Cheng, C., Yan, S., Yang, L., and xiang Jiang, S., 2026, "Three-dimensional printed lattice structures for next generation of wearable technology," *Fashion and Textiles*, Mar 2026.
- [37] Shahriari, B., Swersky, K., Wang, Z., Adams, R. P., and de Freitas, N., 2016, "Taking the Human Out of the Loop: A Review of Bayesian Optimization," *Proceedings of the IEEE*, **104**(1), pp. 148–175.
- [38] Frazier, P. I., 2018, "A Tutorial on Bayesian Optimization," *arXiv preprint arXiv:1807.02811*.
- [39] Ament, S., Daulton, S., Eriksson, D., Balandat, M., and Bakshy, E., 2023, "Unexpected Improvements to Expected Improvement for Bayesian Optimization," *Advances in Neural Information Processing Systems*, Vol. 36, doi: [10.52202/075280-0904](https://arxiv.org/abs/2010.08195).
- [40] Daulton, S., Balandat, M., and Bakshy, E., 2021, "Parallel Bayesian Optimization of Multiple Noisy Objectives with Expected Hypervolume Improvement," *Advances in Neural Information Processing Systems*, Vol. 34, pp. 2187–2200, <https://arxiv.org/abs/2105.08195>
- [41] Daulton, S., Cakmak, S., Balandat, M., Osborne, M. A., Zhou, E., and Bakshy, E., 2022, "Robust Multi-Objective Bayesian Optimization Under Input Noise," *Proceedings of the 39th International Conference on Machine Learning*, Proceedings of Machine Learning Research, Vol. 162, pp. 4831–4866, <https://arxiv.org/abs/2202.07549>
- [42] Low, A. K. Y., Mekki-Berrada, F., Ostudin, A., Xie, J., Vissol-Gaudin, E., Lim, Y.-F., Gupta, A., Li, Q., Ong, Y. S., Khan, S. A., and Hippalgaonkar, K., 2024, "Evolution-guided Bayesian Optimization for Constrained Multi-Objective Optimization in Self-Driving Labs," *npj Computational Materials*, **10**, pp. 1–11.
- [43] Baptista, R. and Poloczek, M., 2018, "Bayesian Optimization of Combinatorial Structures," *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pp. 462–471, <https://proceedings.mlr.press/v80/baptista18a.html>
- [44] Garrido-Merchán, E. C. and Hernández-Lobato, D., 2020, "Dealing with Categorical and Integer-Valued Variables in Bayesian Optimization with Gaussian Processes," *Neurocomputing*, Vol. 380, pp. 20–35, doi: [10.1016/j.neucom.2019.11.004](https://doi.org/10.1016/j.neucom.2019.11.004).
- [45] Perdikaris, P., Raissi, M., Damianou, A., Lawrence, N. D., and Karniadakis, G. E., 2017, "Nonlinear Information Fusion Algorithms for Data-Efficient Multi-Fidelity Modelling," *Proceedings of the Royal Society A*, **473**(2198), p. 20160751.
- [46] International Organization for Standardization, 1999, "ISO 11334-4:1999 Assistive products for walking manipulated by one arm — Requirements and test methods — Part 4: Walking sticks with three or more legs," Standard, International Organization for Standardization, <https://www.iso.org/standard/27298.html>
- [47] Requejo, P. S., Wahl, D. P., Bontrager, E. L., Newsam, C. J., Gronley, J. K., Mulroy, S. J., and Perry, J., 2005, "Upper extremity kinetics during lofstrand crutch-assisted gait," *Medical engineering & physics*, **27** 1:19–29, Jan 2005.
- [48] Rasouli, F. and Reed, K. B., 2020, "Walking assistance using crutches: a state of the art review," *Journal of Biomechanics*, **98**:109489, Jan 2020.
- [49] Manocha, R. H. K., MacGillivray, M. K., Eshraghi, M., and Sawatzky, B. J., 2021, "Injuries associated with crutch use: a narrative review," *PM&R*, **13**:1176–1192, Dec 2021.
- [50] da Silva, D. B.-B. and Fachin-Martins, E., 2022, "Pain mapping and health-related conditions in relation to forearm crutch usage: a cross-sectional study," *Assistive Technology*, **34**:334–340, Oct 2022.
- [51] Dozono, K., Hachisuka, A., Wada, F., and Hachisuka, K., 2015, "Peripheral neuropathies in nonparetic upper extremities of stroke patients induced by excessive use of a walking device," *Journal of stroke and cerebrovascular diseases* : the official journal of National Stroke Association, **24** 8:1841–7, Aug 2015.
- [52] Segura, A. and Piazza, S. J., 2007, "Mechanics of ambulation with standard and spring-loaded crutches," *Archives of physical medicine and rehabilitation*, **88** 9:1159–63, Sep 2007.
- [53] Zhang, Y., Liu, G., Xie, S., and Liger, A., 2011, "Biomechanical evaluation of an innovative spring-loaded axillary crutch design," *Assistive Technology*, **23**:225–231, Dec 2011.

- [54] MacGillivray, M. K., Manocha, R. H., and Sawatzky, B., 2016, "The influence of a polymer damper on swing-through crutch gait biomechanics," [Medical engineering & physics](#), 38 3:275-9, Mar 2016.
- [55] Mottaghi, M., Woods, M. C., Danier, L., So, A., Reeves, J. M., and Pearce, J. M., 2025, "Open-source 3d printable forearm crutch," [PLOS ONE](#), Nov 2025.
- [56] Stasiak-Cieślak, B. and Malawko, P., 2025, "Expert evaluation of attachments (caps) for orthopaedic equipment dedicated to pedestrian road users," [Open Engineering](#), Jan 2025.
- [57] Wang, H.-Y., Wu, L., Qi, J., null, J.-T. D., and Wang, Y., 2025, "Porous lattice structure optimization in 3d printed insole design," [Journal of Fiber Bioengineering and Informatics](#), 18:29-39, Jun 2025.
- [58] Zhang, J., Lu, S., Lin, Y., Wang, Y., Yi, X., and Fang, W., 2024, "Pressure-reducing design of 3d-printed diabetic shoe midsole utilizing auxetic lattice structure," [Applied Sciences](#), 14:5291, Jun 2024.
- [59] Janarthanan, A. K. K. and Vaidhyanathan, B., 2024, "Additive manufacturing of smart footwear components for healthcare applications," [Micromachines](#), 16:30, Dec 2024.

List of Figures

1	Closed-loop, experiment-driven design framework. Parameterized PLA/TPU tensegrity-inspired unit cells are printed, tested under quasi-static compression and drop-weight impact, and the resulting performance data drive a Gaussian-process surrogate that recommends the next batch of designs to fabricate.	2
2	Working T_3 -prism prototype. (a) Parametric OpenSCAD model whose base radius R , cell height H , twist angle θ , strut diameter d_s , and cable diameter d_t are the design variables of Table 1. (b) A representative dual-material print from the PR #35 print history (rigid PLA struts, here in white; tendon members in orange), with callouts marking the strut diameter d_s , a tendon (cable) diameter d_t , and the cell height H ; the inter-plate twist θ is the relative rotation of the top and bottom node triangles, and the base radius R sets their circumradius.	3
3	Fabrication and characterization workflow for the multi-material tensegrity-inspired unit cells: design parameters drive parametric CAD, slicing with manually generated TPU supports, a single multi-material print on the Bambu Lab H2D, post-processing and inspection, and mechanical testing. Node photographs are representative artifacts drawn from the project's build history: the OpenSCAD T_3 -prism render (CAD), the dual-nozzle Bambu Studio slice with PLA on the left extruder and TPU on the right (slicing), an in-progress H2D print of the prism batch (FFF), an as-printed specimen (post-processing), and the bungee-assisted drop-tower base plate with accelerometer (testing).	4
4	<i>Illustrative example (synthetic data)</i> . Template for the mechanism-oriented drop-test figures planned for the Results: (a) processed deceleration response of a rigid control vs. a tensegrity specimen (SAE J211 CFC-180 filtered), with event-based phase shading and numbered markers keyed to the schematic frames (b)–(d); (a2) cumulative-impulse subpanel showing the two traces transfer the same mass-normalized Δv , so the lower tensegrity peak reflects temporal redistribution rather than inconsistent loading; (e) the internal-anchor joint load path. Curves are synthetic and impulse-consistent (control anchored to the documented ~ 1792 G CFC-180 peak; tensegrity shoulder scaled to the same impulse), pending real high-rate channels and camera frames. Generated by <code>scripts/figures/mechanistic_data_figure_example.py</code>	5
5	<i>Illustrative example (synthetic data)</i> . Leave-one-out cross-validation of the Gaussian-process surrogate—predicted vs. observed specific energy absorption (SEA) and peak transmitted force, with model error bars and R^2_{LOO} —obtained from a real Ax/BoTorch run on synthetic objective values. To be regenerated from campaign data.	5
6	<i>Illustrative example (synthetic data)</i> . Parameter-sensitivity ranking (model length-scale-based importances per objective; cf. SAASBO inverse length-scales or Sobol indices) of the design variables in Table 1, from the same Ax/BoTorch run as Fig. 5.	6
7	<i>Illustrative example (synthetic data)</i> . Best-so-far SEA vs. number of physical experiments for the Ax/BoTorch loop against an independent random-search baseline; the shaded region is the $n = 9$ Sobol initialization. Generated on synthetic objective values; to be replaced with campaign results (with confidence intervals over random seeds, and Pareto hypervolume for the multi-objective runs).	6
8	<i>Illustrative example (synthetic data)</i> . Pareto front in (peak transmitted force, SEA) space with all evaluated designs, from the Ax/BoTorch loop on synthetic objective values; representative geometries to be annotated once campaign data are available.	6

List of Tables

1	Design variables for the T_3 -prism Sobol initialization batch, as implemented in the BO harness (single-iteration $n = 9$ Sobol set). The joint diameter is held fixed at the <code>t3-prism.scad</code> default; all specimens are printed in the vertical orientation and packed 3×3 on a single Bambu Lab H2D plate.	3
2	Multi-material print parameters on the Bambu Lab H2D. Cells marked <i>TBD</i> are pending confirmation; the reporting structure follows the methods section of Ye et al. [12].	4
3	<i>Placeholder</i> . Mean and CV of SEA, peak force, and compaction efficiency across n replicate prints for the top- k designs.	6